

Bhagyesh Rath

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EDUCATION

San Jose State University

Master of Science in Artificial Intelligence

San Jose, CA

Expected: May 2027

Relevant coursework: Machine Learning, Deep Learning, AI and Data Engineering, Natural Language Processing, AI Threat Intelligence

San Jose State University

Bachelor of Science in Computer Science, Magna Cum Laude

San Jose, CA

August 2020 - May 2024

Relevant coursework: Java OOP, Data Structures and Algorithms, Object-Oriented Design, Software Engineering, Operating Systems, Relational Databases (SQL), Machine Learning, Advanced Programming in Python, Data Visualization, Information Security.

PUBLICATIONS

Rathi, B.*, Chawla, E.*, Ershov, A., Andreopoulos, W. B. "A Comparative Evaluation of Retrieval Pipelines for Large-Scale Scientific Question Answering with Open-Weight LLMs." IEEE AIXSET 2026 (in press). (* Equal contribution) | [Code](#)

RELEVANT EXPERIENCE

Scalar Field (YC P25) - Forward Deployed Engineer Intern | Seattle, WA

June 2026 – August 2026

- Built a HubSpot MCP (Model Context Protocol) server exposing CRM data and actions as tools for LLM agents, enabling automated GTM workflows over the company's customer data.
- Developed a RAG chatbot that retrieves lead information from a knowledge base to answer GTM queries, handling retrieval and structured responses over internal data.

GitLab - Open Source Contributor | Remote

April 2026 – Present

- Contributed a security fix to the GitLab monorepo enforcing streaming JSON validation limits in the import pipeline's NdjsonReader to mitigate DoS from malicious export archives ([MR !224828](#)).
- Added a REST DELETE endpoint to GitLab's Package Registry with authorization policies, request specs, and API docs, aligning behavior with existing upload/download endpoints ([MR !242461](#)).

Rakuten - Software Engineer Co-op | San Mateo, CA

August 2023 – December 2023

- Developed event listeners in Kotlin to handle consent-revoked and delete-account events from Google, Facebook and Apple.
- Implemented features to validate and decode JWTs to extract required data, and deserialize JSON responses.
- Built an SDK to abstract low level service to service calls, published artifact in Artifactory through GitLab CI/CD.
- Achieved 85% test coverage (JaCoCo) with Unit, Integration testing, and leveraged SonarQube to eliminate code smells.
- Deployed microservices on Google Cloud Platform (GCP) using GitLab CI/CD automated build, test, and deploy pipelines.
- Created Dockerfiles and orchestrated containers using Kubernetes, deploying across non-prod and prod environments.
- Integrated OpenTelemetry and Google Cloud Trace for real time monitoring and performance analysis.

Rakuten - Software Engineer Intern | San Mateo, CA

May 2023 – August 2023

- Contributed to Social Sign Up / Login with Google, Apple, Facebook for Rakuten businesses in Spain, UK, and Germany.
- Achieved a 40% adoption rate of Social Authentication and a 37% increase in CVR over Username and Password Authentication.
- Developed Microservice in Kotlin for social link (with Google, Facebook, Apple) management (Display and Unlink) features.
- Implemented REST APIs leveraging OpenID Connect & OAuth2.0 to implement token based Authentication and Authorization.
- Implemented structured logging using Log4j, analyzed the logs through GCP Cloud Monitoring.
- Participated in Agile software development such as daily stand-ups, sprint planning, and sprint review.

San Jose State University - Teaching Assistant, CS Department | San Jose, CA

August 2022 – December 2022

- Conducted a 3-hour weekly lab to help 50+ students apply OOP concepts in Java and develop object-oriented thinking.
- Conducted weekly office hours to solve homework or lab-related problems and helped students over email.
- Worked closely with CS department staff and multiple professors to design and improve the lab.

RESEARCH

RAG Evaluation Framework for Scientific QA | Python, SPECTER2, ChromaDB, ColBERT, Ollama | [Code](#)

June 2026

- Built a reproducible, config-driven pipeline (Python) comparing six retrieval strategies for scientific question answering over a ~460K-paper arXiv corpus, from a research prototype into a modular, release-ready codebase.
- Implemented SPECTER2 domain embeddings with task-specific document/query adapters, a ChromaDB vector store, and an agentic tool-calling retriever via function-calling APIs.
- Deployed and optimized the full stack on an NVIDIA DGX Spark (GB10, ARM64/CUDA 13), including getting ColBERT's PLAID late-interaction index compiling on a novel architecture and serving open-weight LLMs (Llama-3.1, Qwen2.5) locally via Ollama.

- Designed an LLM-as-a-judge evaluation with an explicit answer/refusal gate and both conditional and unconditional scoring to fairly compare strategies with differing answer rates.
- Parallelized generation and evaluation stages for a $\sim 6\times$ throughput improvement across $\sim 20K$ queries.
- Published as a co-first-author paper at IEEE AIXSET 2026 (in press).

PathSafe Agent: Does an Agent's Safety Depend on Where a Harmful Instruction Comes From? (*In progress*)

PROJECTS

RAG-Powered Interactive Portfolio | Next.js, Pinecone, Vertex AI, Vercel AI SDK, LangChain | [GitHub](#), [Chat](#) Feb 2026

- Shipped a production-grade RAG chatbot serving real users on a live portfolio site with full ingestion pipeline.
- PDF ingestion $\rightarrow$ recursive chunking $\rightarrow$ 768-dim embeddings $\rightarrow$ Pinecone upsert, with top-K retrieval, score thresholding, and metadata filtering to minimize hallucination.
- Built a streaming API endpoint using Vercel AI SDK with Gemini 2.5 Flash, achieving sub-second TTFT.
- Implemented multi-layer security: input validation, system prompt hardening against prompt injection, and topic-scoped guardrails.
- Authored a 27-test integration suite covering embedding quality, retrieval accuracy, pipeline correctness, and injection defense.

SRE Copilot: AI Incident Triage Agent for On-Call SREs | Python, PyTorch, LangChain, ChromaDB | [GitHub](#), [Colab](#) Feb 2025

- Built a ReAct tool-use agent that triages production incidents across 5 data sources (RAG over 768 runbooks, text-to-SQL, GCP status API, vulnerability DB, web search) using dual Qwen 2.5 models (32B + 3B) on a single GPU via 4-bit NF4 quantization.
- Evaluated 4 prompting techniques (baseline, prompt chaining, self-reflection, meta-prompting) across 80 benchmarks, finding self-reflection improved answer grounding by 125% (0.40 $\rightarrow$ 0.90) and actionability by 100% (0.50 $\rightarrow$ 1.0) over baseline, while prompt chaining achieved perfect tool selection on the 32B model but failed on 3B — revealing minimum capability thresholds for structured data.
- Achieved $\sim 4\%$ prefill speedup on the 32B model by building KV cache reuse for system prompt acceleration.
- Implemented defense-in-depth security (SQL injection blocking, prompt hardening) passing 3/5 red-team attacks per model; deployed via Gradio with real-time model/technique comparison.

Caption Lens: Visual-Semantic Alignment via Attention Mechanisms | PyTorch, Hugging Face | [App](#) / [Repo](#) May 2025

- Designed and trained three encoder-decoder image captioning models (Show-and-Tell, Show-Attend-and-Tell with soft attention, Visual Sentinel adaptive attention) on MS COCO 2014 (82K images, $\sim 9K$ -word vocab) using a fine-tuned ResNet-101 encoder feeding $7\times 7\times 2048$ spatial features into an LSTM decoder.
- Implemented Bahdanau soft attention with doubly stochastic regularization, boosting CIDEr from 1.023 $\rightarrow$ 1.040 and winning all 8 metrics (BLEU-1–4, METEOR, ROUGE-L, CIDEr, SPICE); BLEU-4 of 0.329 exceeds the original 2015 paper (0.250).
- Engineered beam search with 1–10 beam-width ablation lifting CIDEr from 1.013 (greedy) to 1.093 (beam=6); built interpretable attention heatmaps tracing each word to its driving image region.
- Deployed as an interactive multi-model demo on Hugging Face Spaces via Docker + Gradio with real-time model switching.

AutoML Studio: Full-Stack Platform for Classification & Regression with Code Generation | Python, Flask, scikit-learn, XGBoost, React 18 | [GitHub](#) Dec 2024

- Built a dual-mode AutoML platform that auto-detects classification vs regression and trains 6 classification models (LogReg, RF, SVM, KNN, XGBoost, DT) and 5 regression models in parallel via Flask background threads, exporting the winner as both a serialized .pkl and a standalone reproducible Python script — eliminating vendor lock-in vs Google AutoML / H2O.
- Engineered a hardware-aware leaderboard by spawning a psutil background thread sampling RAM/CPU every 200 ms during each training run, enabling sort-by-memory-footprint alongside accuracy and training time — a production-budget signal that accuracy-only AutoML tools ignore.
- Optimized for scale with adaptive RandomizedSearchCV iteration budgets that grow with parameter-grid size, automatic SVM $\rightarrow$ SGD substitution on $>2K$ -row datasets to avoid $O(n^2)$ training blowup, stratified sampling that preserves class ratios on downsampled data, and a chunk-based CSV reader handling 1 GB+ files without OOM.
- Implemented imbalance-aware training (auto class_weight='balanced' for sklearn, dynamic scale_pos_weight for XGBoost) and deployed a React 18 frontend with worker-based PapaParse for non-blocking uploads, Recharts ROC / confusion-matrix / regression-fit visualizations, and a glassmorphism UI streaming live training logs from the backend.

Skin Cancer Detection using CNN | Python, TensorFlow, Scikit-Learn, Pandas, Seaborn | [Colab](#) June 2024

- Categorized 10,015 skin lesion images into 7 cancer classes, visualizing data with Seaborn for better understanding.
- Preprocessed data through image resizing, normalization, one-hot encoding, and data augmentation.
- Performed minority class oversampling to address class imbalance.
- Developed and implemented a Deep Convolutional Neural Network (CNN) with multiple layers.
- Conducted hyperparameter tuning, created train-test-validation splits, and employed Adam optimizer and learning rate annealer to optimize model performance, achieving 80% accuracy.

Bank Churn Data Analysis and Prediction using ML | Pandas, Seaborn, Matplotlib, Scikit-Learn | [Colab](#) Feb 2024

- Analyzed data of 10,000 account holders at a Multinational Bank by doing exploratory data analysis with Pandas.

- Visualized data using Seaborn and Matplotlib to choose strategies for developing ML training pipelines.
- Implemented data preprocessing, feature selection, established test and training sets, and constructed a streamlined pipeline for training 5 machine learning models to predict customer churn using Scikit-Learn.
- Implemented XGBoost, Random Forest, KNN, SVM, and Naive Bayes Machine Learning models.
- Utilized N-fold cross-validation, F1-score, confusion metrics to evaluate the performance of each model.
- Tested the ML models using a test set to show the performance of the models and compare them to get the most accurate model.

SKILLS & CERTIFICATIONS

Programming Languages: Java, Python, JavaScript, Kotlin, SQL

Frameworks: Flask, Scikit-Learn, TensorFlow, PyTorch, Pandas, NumPy, Seaborn, LangChain

AI/ML Tools & Frameworks: Vertex AI, Google ADK, HuggingFace, LangChain, Ollama, Model Context Protocol (MCP), Claude Code, Gemini, Copilot, Cursor

Skills: Distributed Systems, Multi-tiered Systems, Agile software development, Unit/Integration Testing, Monitoring and Alerting, RAG, LLM Agents (ReAct), LLM Evaluation.

Databases: MongoDB, MySQL, PostgreSQL, Redis, Pinecone, ChromaDB

Tools: GitLab, Artifactory, OpenTelemetry, Google Cloud Trace

Cloud: GCP, Docker, Kubernetes, AWS, Vercel

Certifications: [Machine Learning Specialization \(Stanford\)](#)